

An integrated perception, navigation, and manipulation pipeline for semi-autonomous door opening with a mobile manipulator

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Abstract—Autonomous door opening is a key capability for service robots. It requires integration of perception, navigation, localization, and manipulation. This paper presents a semi-autonomous door-opening pipeline for a mobile manipulator. The system requires manual initialization of the robot pose in Rviz; it autonomously detects and approaches the door, estimates the 3D handle position, aligns the mobile base, performs the pre-grasp positioning, and executes handle manipulation and door opening. The system was evaluated on a real mobile manipulator in 20 trials: 10 nominal operation trials, 5 trials with varying initial robot-to-door distance, and 5 trials varying initial end effector height. In nominal condition, door perception, navigation, 3D localization, alignment and pre-grasp positioning were successful 100% of the time, while the complete door opening succeeds 80% of the time. The system successfully handled the initial robot-to-door distance in four of five trials. However, changes in the initial end effector height reduced performance, completing only one of five trials. Failures occurred mainly in the contact and interaction between the handle and the gripper. Physical handle manipulation is the main limitation of the current system.

I. INTRODUCTION

Service robots operating in human-centered environments, must combine autonomous mobility with the ability to physically interact with infrastructure originally designed for humans. Among these interactions, door opening is particularly relevant because doors represent unavoidable barriers between indoor spaces. A robot unable to autonomously detect, approach or manipulate a door, may therefore require human assistance to continue its task, limiting the robot autonomy. Previous robotic systems have demonstrated autonomous or semi-autonomous door interaction using personal robots, rescue robots and mobile manipulators [1]–[3]. Autonomous door opening in robotics systems is a challenging task that requires a combination of several tightly coupled capabilities, including estimation of door and handle pose, robot positioning and configuration of the robot approach, and physical interaction with the door through a stable grasp and physically actuation of the door mechanism. Different

approaches have been proposed to tackle subproblems of this task. Some works focus on semantic keypoint-based handle localization [4], 3D door pose estimation and robot positioning [5], force estimation [6], or Cartesian impedance control [7], while others employ more advanced algorithms for adaptive position, force control that utilize reinforcement learning to further improve mobile manipulator interaction for the challenging scenario of different door configurations [8]. These studies demonstrate that successful door opening depends not only on perception or manipulation individually, but also on the coordination between the different stages of the task.

The external forces acting on the robot while opening the door can come from many sources, such as the robot configuration, relative positioning or sequencing of the individual stages. For example, [9] and [6] report that the configuration and positioning of the robot have a significant influence on the forces generated when opening a door. Evaluating only a binary outcome of the task (door opened or not), does not provide information in which stage a failure occurred. In fact, evaluating the performance based on the different stages of the task has been shown to provide more diagnostic evidence and more informative measures than simply evaluating the success of the task as a binary metric [10]. Therefore, the presented work evaluates the pipeline at a stage-level.

In this paper, we present a stage-level evaluation of an integrated perception, navigation and manipulation pipeline for opening a door with a semi-autonomous mobile manipulator. The system requires manual initialization of the robot pose in Rviz. After this, the pipeline autonomously searches for the door, uses RGB-D for perception of the scene to navigate towards it, and localize the handle in 3D using point-cloud data. Then, the mobile-base aligns to the door before the robot performs physical interaction to open it. We implemented this pipeline in ROS2 [11] and carried out the evaluation by performing the overall task starting from different initial robot-to-door distances and with different initial end-effector heights. We carried out this evaluation for 20 trials on the physical robot: 10 nominal operating conditions, defined by an initial robot-to-door distance of 0.35 m and an initial end effector height of 0.44 m, five trials varying the robot-to-door distance and five trials varying initial end effector height. The main contributions of this work are:

- An integrated perception–navigation–manipulation pipeline that coordinates door detection, autonomous approach, 3D handle localization, fine alignment, and

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robotic manipulation for door opening with autonomous execution after manual pose initialization.

- A stage-by-stage experimental evaluation that characterizes the performance of the complete pipeline from door perception to physical door interaction, enabling the identification of the stages responsible for task failures.
- An experimental evaluation based on 20 real-robot trials under nominal and varying initial configurations, including robot-to-door distance and end-effector height, to investigate their influence on end-to-end performance.

II. RELATED WORK

A. Autonomous door opening

A major task for personal and service robots is autonomous door opening. Meeussen et al. demonstrated an early integrated door-opening system on the PR2 platform, combining navigation, vision, laser sensing, and compliant manipulation [1]. More recent systems have investigated door interaction using assistant and rescue robots [2], [3]. Several authors have addressed different challenges related to door opening task, while Mitterhuber et al. studied RGB-D-based door pose estimation and robot positioning [5], Wang et al. presented semantic keypoint representations to infer interaction points for manipulation from visual observations [4]. Physical interaction has been the subject of other studies: Nishimura et al. used force estimation to manipulate doors [6], and Stuede et al. showed how to use Cartesian impedance control to open and traverse doors [7]. To handle uncertainty during door interaction, Kang et al. further integrated reinforcement learning with adaptive position–force control [8]. The significance of the robot configuration during physical interaction was demonstrated by Fukumoto and Jinnai’s investigation of coordinated locomotion and manipulation for door opening and passing [9]. These studies show significant advancements in the field of autonomous door contact. Errors in perception, placement, or localization can affect subsequent gripping and physical engagement, but door opening is still a system-level issue. Therefore, examining intermediate execution phases is helpful for determining the variables limiting system reliability in addition to end-to-end job success.

Recent work has investigated different door handling systems under different conditions. Xiong et al. [12] proposed a full-stack mobile manipulation system that can operate on doors and other articulated objects in open-world environments. In addition to perceiving these objects using RGB-D data, their system supports physical interaction through parameterized grasping and manipulation primitives. The system models initial interactions using behavior cloning and then refines them via online reinforcement learning on previously unseen objects. The authors evaluated their approach on 20 articulated objects in four different buildings, reporting a success rate of 50% without adaptation, which improved to 95% after online adaptation. An alternative approach to door handling was illustrated by Kang et al. [8] where perception and interaction with a door were conducted via point-cloud processing and position–force control with

reinforcement learning. Schulze et al. [13] integrated door interactions into the long-term deployment of a mobile service robot, which included elevator interactions, and evaluated the variability with regard to the robot-to-door positioning. Aspects of these approaches motivate our approach. However, our work differs in evaluating performance on a pipeline of door-opening tasks across seven discrete stages, comparing performance under controlled variations in initial robot-to-door distance and initial end-effector height.

B. Integrated perception and mobile manipulation

Mobile manipulation aims to leverage autonomous navigation and perception to perform manipulation tasks in locations beyond the workspace of a fixed manipulator. Integration of robotic software frameworks such as ROS2 [11], with the Nav2 [14] navigation stack and the MoveIt [15] motion-planning libraries provide a solid base for achieving this goal. Consistent spatial information as well as output from perception modules, can be used to position a mobile base and guide manipulation, such as door opening based on an estimated handle pose. Leveraging depth from RGB-D sensing allows geometric estimations, such as a door frame, to be improved for accurate manipulation. This can involve using a point-cloud processing technique, such as RANSAC [16], implemented in the Point Cloud Library (PCL) [17]. Previous work on door perception is based on using geometric or color information from 3D sensing [18].

In this paper we describe a pipeline to open doors with a mobile manipulator in ROS2. This paper does not contribute a new perception or control algorithm, but rather a staged (and characterizable) integration of perception and control. We evaluate seven stages of the integrated pipeline externally and characterize end-to-end task success as a function of the initial robot-to-door distance and initial end-effector height. This evaluation allows to analyze the sensitivity of certain failures to specific stages of the pipeline. In contrast to existing studies, we focus on identifying failure stages with controlled variations in the initial robot configuration rather than adaptability to different door configurations or long-term operation.

III. PROPOSED SYSTEM

For semi-autonomous door opening, the proposed system integrates vision, navigation, 3D localization, base alignment and robotic manipulation into a sequential ROS2 based pipeline. Before execution, the operator, has to initialize the robot pose on the map using Rviz. After this, the system automatically executes the seven stages, starting from searching for the door and ending with grasping the handle and attempting to open the door. Each stage outputs the spatial information required by the next stage for navigation or manipulation.

A. Mobile Manipulator Platform

The system was implemented on the CALL-M robot mobile manipulator [19], [20], developed at CNRS-AIST JRL for indoor robotics tasks. The platform consists of a TriOrb

omnidirectional mobile base, a Universal Robots UR7e six-degrees-of-freedom collaborative manipulator, and a Robotiq 2F-85 parallel gripper. Two RPLidar A1M8 laser scanners are used for navigation and obstacle detection, while an Intel RealSense D435i RGB-D camera mounted near the end effector provides color and depth information for door handle perception. Computation is distributed between a Jetson Orin Nano, used for sensor processing and perception; and an Intel NUC, for manipulation and navigation control. Both computers run Ubuntu and ROS2 Humble communicating through the same ROS2 network. The hardware configuration is shown in Fig. 1.

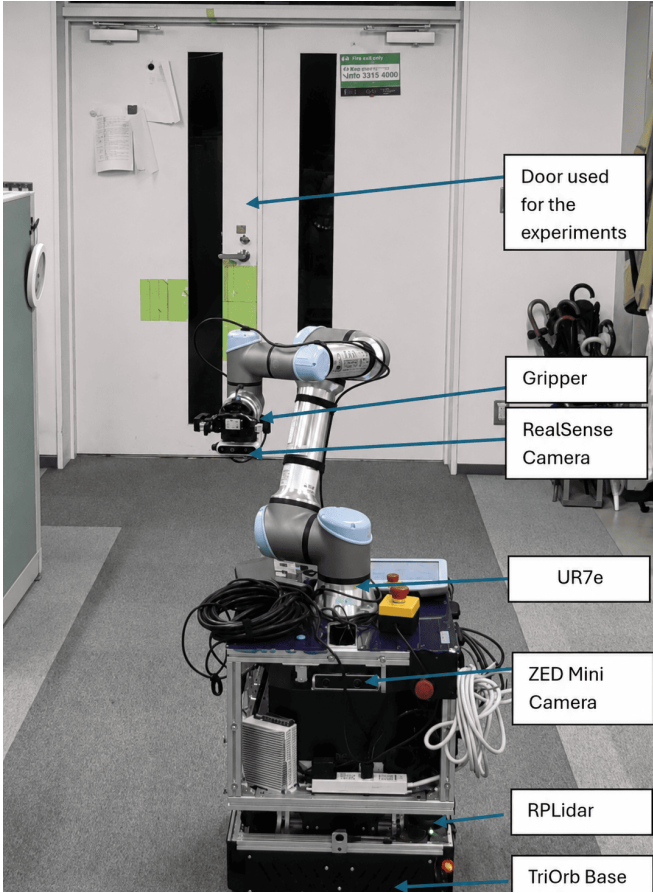


Fig. 1. CALL-M mobile manipulator and main hardware components used in the door-opening experiments.

B. System Architecture

The perception, navigation and manipulation modules are interconnected via the ROS2 middleware. The perception module processes RGB-D data to detect the door and the handle and estimates the 3D handle position based on depth and point-cloud data. Then, a transformation is performed to retrieve the point data in the camera, manipulator, mobile-base and map reference frame using the ROS2 transformation system [21].

The autonomous positioning of the robot/mobile-base is achieved by using the ROS2 navigation framework [14] to handle localization, path planning, obstacle avoidance, and

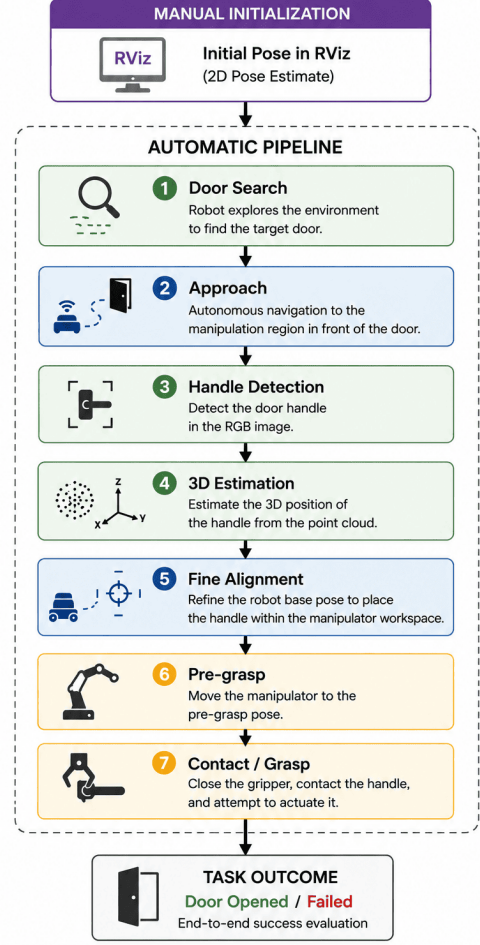


Fig. 2. Integrated door-opening pipeline. The robot pose is manually initialized in RViz before execution. The subsequent pipeline is executed automatically and comprises perception (green), navigation and base positioning (blue), and manipulation (yellow) stages.

trajectory execution. Once in the manipulation region, the estimated handle pose is used to refine the alignment of the mobile-base to the manipulation target followed by planning and executing arm motion using the MoveIt manipulation framework [15].

C. Semi-Autonomous Door-Opening Pipeline

The door opening procedure begins with a manual initialization step, followed by seven stages automatically executed, as shown in Fig. 2. During initialization, the operator manually specifies the pose of the robot in the generated map using RViz. The initial pose is an essential input to the ROS2 navigation system’s localization mechanism, expressed as an estimate of the robot’s 2D pose. This initializes the robot’s 2D pose estimate and this information is required before autonomous navigation begins. The complete execution sequence of the proposed door-opening pipeline is illustrated in Fig. 3.

In the first stage, The automatically executed pipeline starts with the stage *Door Search*, in which the mobile base rotates around its vertical axis while the perception

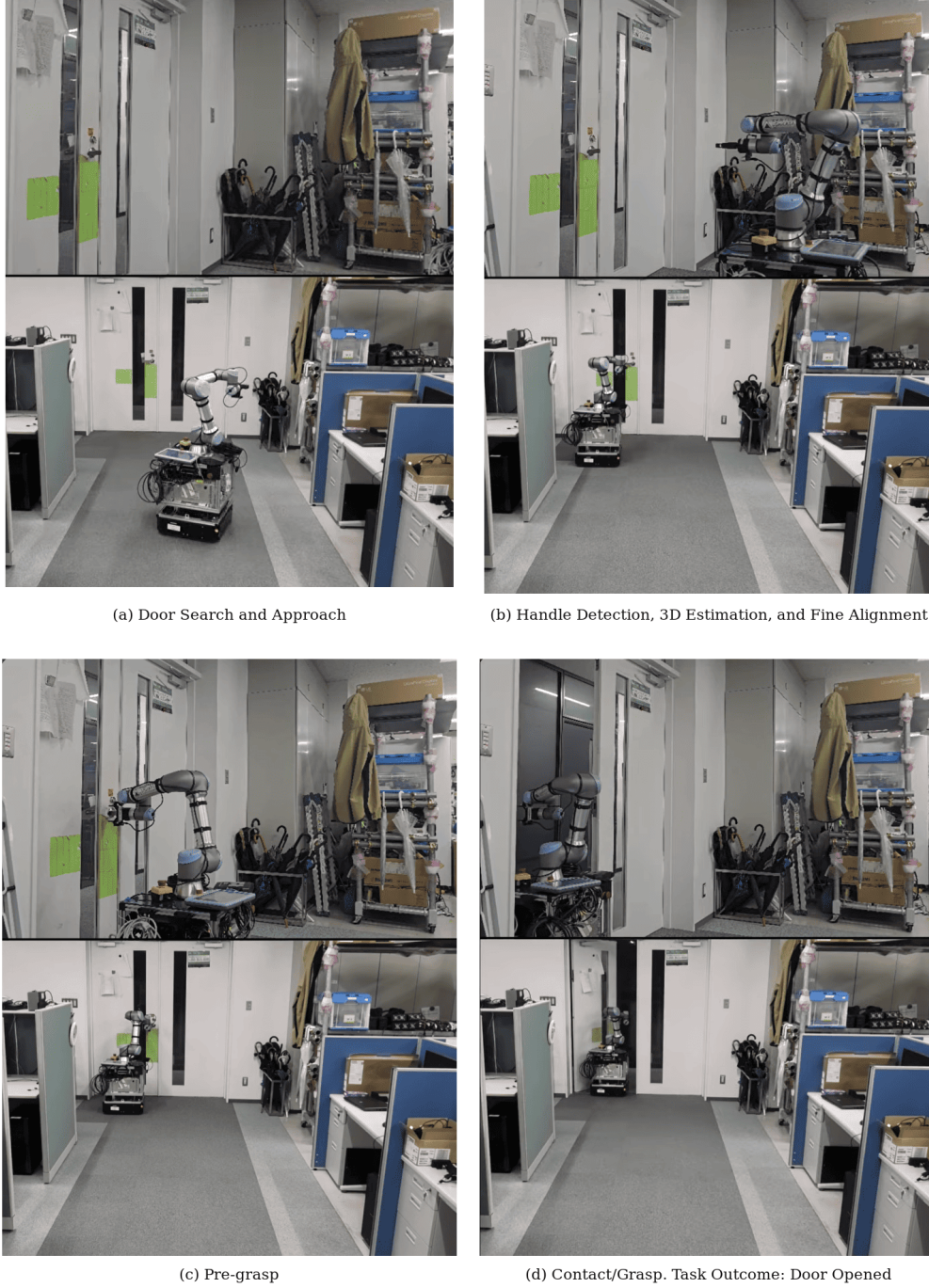


Fig. 3. Execution sequence of the proposed door-opening pipeline: (a) door search and approach; (b) handle detection, 3D estimation, and fine alignment; (c) pre-grasp positioning; and (d) contact and grasp. The final task outcome is a successfully opening of the door.

module processes RGB-D observations from a ZED mini camera to detect the door and estimate its distance. The rotation is performed until the target door is detected in the sensor field of view. Door detection was done by using a YOLO model trained with a total of 4656 images, 3000 from DeepDoors2 [22] and 1656 collected locally, divided in 3893 training images and 743 images for validation. The dataset also included negative indoor images without doors to reduce

false positive detections.

In the second stage, *Approach*, the robot uses the detected door position as a target and autonomously navigates towards it using the ROS2 navigation framework. The base approaches the door until stopping at an approximate distance of 0.80 m, ensuring enough space for subsequent base alignment and manipulator positioning.

In the third stage, *Handle Detection*, the perception mod-

ule analyzes the RGB image taken closer to the door to detect the handle and obtain its 2D image coordinates. Additionally, information about the detected door and handle is used to further correct the orientation of the mobile base with respect to the door and its lateral position, with the goal of positioning the robot in front of the handle.

In the fourth stage, *3D Estimation*, the detected 2D handle coordinates are used, along with the depth and point-cloud data produced by the RGB-D camera, to estimate the handle position in 3D. This estimated point, originally in the camera reference frame, is converted to the manipulator, mobile-base and map reference frames using the ROS2 transformation system. The robot can then advance toward the estimated handle position until it reaches an appropriate distance for manipulation.

The fifth stage, *Fine Alignment*, consists of small corrective motions executed by the mobile base to position the handle at the desired location relative to the robot. These motions place the handle within the manipulator workspace and align the end effector along the estimated height and lateral position of the handle.

In the sixth stage, *Pre-grasp*, the manipulator moves from its initial configuration to a pre-grasp pose. The gripper is oriented according to the handle configuration, and the end-effector is positioned along the handle position, maintaining a suitable offset to avoid interference with objects.

Once the pre-grasp pose is reached, the manipulator executes the final *Contact/Grasp* stage. First, the end effector advances from the pre-grasp pose along the predefined approach direction until the gripper surrounds the handle. The gripper then closes to secure the handle. While the mobile base remains stationary, the manipulator follows a predefined Cartesian trajectory that moves the end effector downward to rotate the lever handle and disengage the latch. The manipulator subsequently pulls the grasped handle toward the robot, causing the door leaf to rotate about its hinges. This motion is executed without online pose correction during physical interaction. A successful *Contact/Grasp* stage indicates that the gripper secured and actuated the handle. End-to-end success was evaluated separately and required the latch to disengage and the door leaf to move to the predefined open position. Door-opening success was not detected automatically by the robot; it was determined after each trial through direct observation and inspection of the recorded video. A trial was classified as unsuccessful when the handle was not securely grasped, the safety force limit was triggered, the latch remained engaged, or the door leaf did not reach the required open position.

A successful trial video showing manual initialization operation and the seven stages of automatically executed door-opening pipeline can be found in the supplementary material.

IV. EXPERIMENTAL SETUP

A. Experimental Environment and Procedure

The system was evaluated in a simple indoor laboratory environment, consisting of a corridor and a hinged door

equipped with horizontal lever handle, as shown in Fig. 1. All experiments were performed with the mobile manipulator CALL-M introduced in Section III. Before starting an experiment, we ensured similar illumination conditions and static environment. Before each experiment, the initial pose of the robot was manually initialized in RViz using the 2D Pose Estimate tool. This step was required to initialize the global pose estimate the ROS2 localization system, since this work did not cover automatic global localization. After this, the operator did not further intervene in the execution of the experiment. The robot performed the complete pipeline, including door search and detection, navigation to the manipulation region, handle detection and 3D localization, fine base alignment and pre-grasp positioning; then, it performed the grasp and handle actuation to open the door. We could then easily identify the source of unsuccessful executions. In the *Contact/Grasp* stage, the manipulator stopped during one of the first two nominal trials when the measured force reached the configured safety threshold of 50 N. After the second nominal trial, the safety force threshold was increased to a value above 50 N, and this new setting was used for nominal trials 3–10 and the distance- and height-variation trials. No further force-triggered stops were encountered after this change, so this manipulation should be kept in mind when comparing the first two nominal trials with the rest of the experiments.

B. Experimental Scenarios

Three scenarios were studied on the physical robot within a total of 20 trials. The first scenario is the nominal condition, where we perform 10 trials to test the repeatability of the entire pipeline. The nominal configuration used an initial robot-to-door distance of 0.35m and an initial end effector height of 0.44 m.

In the second scenario, we vary the initial distance from the robot to the door, using the following distances: 2.15, 3.50, 5.90, 7.90 and 10.10 m for a total of five trials. Fig. 4 displays the starting setups used in this scenario.

Finally, in the third scenario we vary the initial height of the end-effector to the following: 0.34, 0.38, 0.44, 0.57 and 0.68 meters for a total of five trials. Fig. 5 displays the basic setups used in this scenario.

C. Evaluation metrics

Performance was measured at both stage and end-to-end levels. A stage was considered successful if it was performed as expected, no failure was triggered, and no intervention from the operator was deemed necessary. The evaluated stages were: (1) door search, (2) approach, (3) handle detection, (4) 3D estimation, (5) fine alignment, (6) pre-grasp, and (7) contact/grasp. Door opening was evaluated separately as the end-to-end task outcome.

For each stage i , the success rate is:

$$SR_i = \frac{N_{\text{success},i}}{N_{\text{trials}}} \times 100\%, \quad (1)$$

where $N_{\text{success},i}$ is the number of successful executions of stage i and N_{trials} is the number of trials in the corresponding

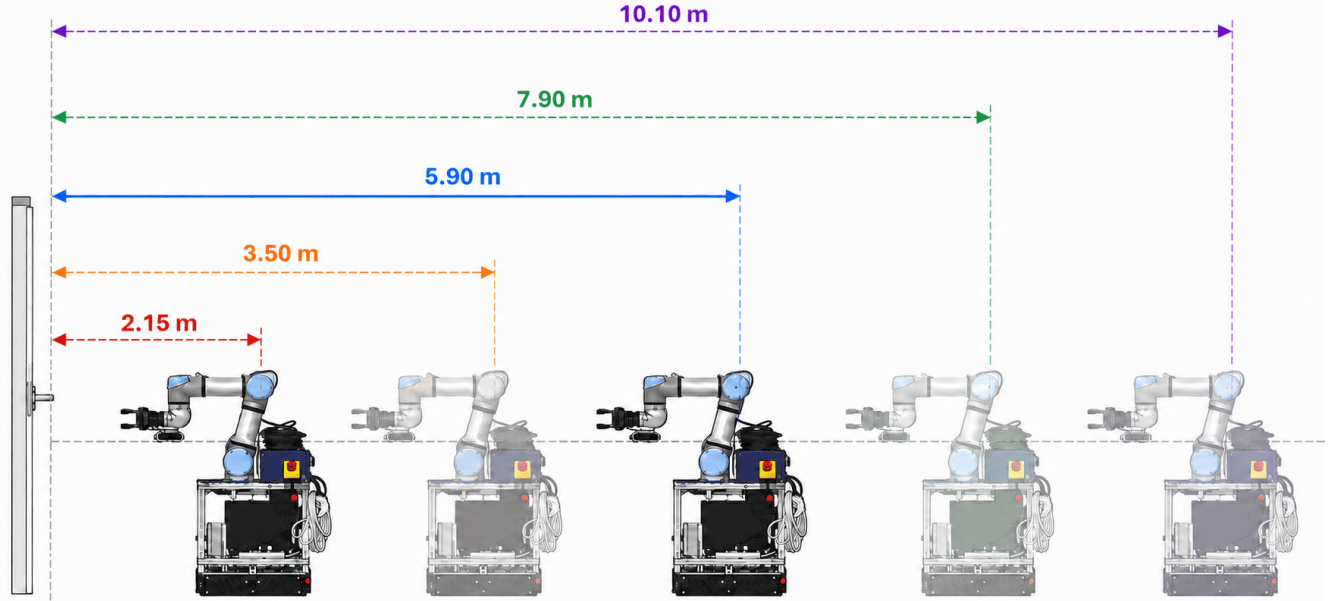


Fig. 4. Initial robot configurations used in the robot-to-door distance-variation trials. The mobile manipulator was positioned at 2.15, 3.50, 5.90, 7.90, and 10.10 m from the door to evaluate the robustness of the door-search and autonomous-approach stages.

scenario. End-to-end success was defined as completion of the complete door-opening sequence. For each unsuccessful trial, we recorded the stage at which the pipeline stopped and classified the immediate cause as navigation/localization failure, geometric misalignment, safety force-limit activation, or insufficient handle actuation to disengage the latch. These failure modes are analyzed in Section V-C.

V. RESULTS AND DISCUSSION

A. Nominal Performance

Under nominal conditions, the robot was able to open the door in 8 out of 10 trials for an end-to-end success rate of 80%. Details of the results are summarized in Table I. The trials for door search, navigation, handle detection, 3D localization, fine alignment, and pre-grasp positioning were successful in each of the trials. The two unsuccessful executions occurred while physically interacting with the handle. In the first trial, the motion of the UR7e was stopped due to the safety system triggering when the interaction force exceeded the configured threshold. The second trial failed because of an approximate 2 cm positioning offset reduced the grasp depth and the force required to actuate the handle, thereby triggering the safety limit. Three recoverable events were recorded during the nominal trials: Two cases resulted from pausing the system with the door at the edge of the camera's field of view causing return to search behavior and one case was due to a repeated re-alignment attempt resulting from outdated point-cloud data during fine alignment stage. These non-failure events were not considered trial failures because the system recovered from them on its own and carried on with the pipeline without operator involvement.

TABLE I
STAGE-LEVEL PERFORMANCE UNDER NOMINAL CONDITIONS

Pipeline stage	Success	Rate
Door search	10/10	100%
Approach	10/10	100%
Handle detection	10/10	100%
3D estimation	10/10	100%
Fine alignment	10/10	100%
Pre-grasp	10/10	100%
Contact / grasp	8/10	80%
End-to-end task	8/10	80%

TABLE II
STAGE-LEVEL PERFORMANCE UNDER DISTANCE VARIATION

Stage	2.15 m	3.50 m	5.90 m	7.90 m	10.10 m
Door search	S	S	S	S	S
Approach	S	S	S	S	F
Handle detection	S	S	S	S	N/A
3D estimation	S	S	S	S	N/A
Fine alignment	S	S	S	S	N/A
Pre-grasp	S	S	S	S	N/A
Contact / grasp	S	S	S	S	N/A
End-to-end task	S	S	S	S	F

S = success; F = failure; N/A = stage not reached because the preceding stage failed.

B. Performance under initial configuration variations

Results from trials involving variations in distance and end-effector height are summarized in Table II. The robot completed the door-opening task starting from distances 2.15 to 7.90 m, but failed to approach the door from 10.10 m due to navigation drift into the boundary of the global costmap.

The nominal height of 0.44 m was the only height at which the end-effector could end-to-end execute the se-

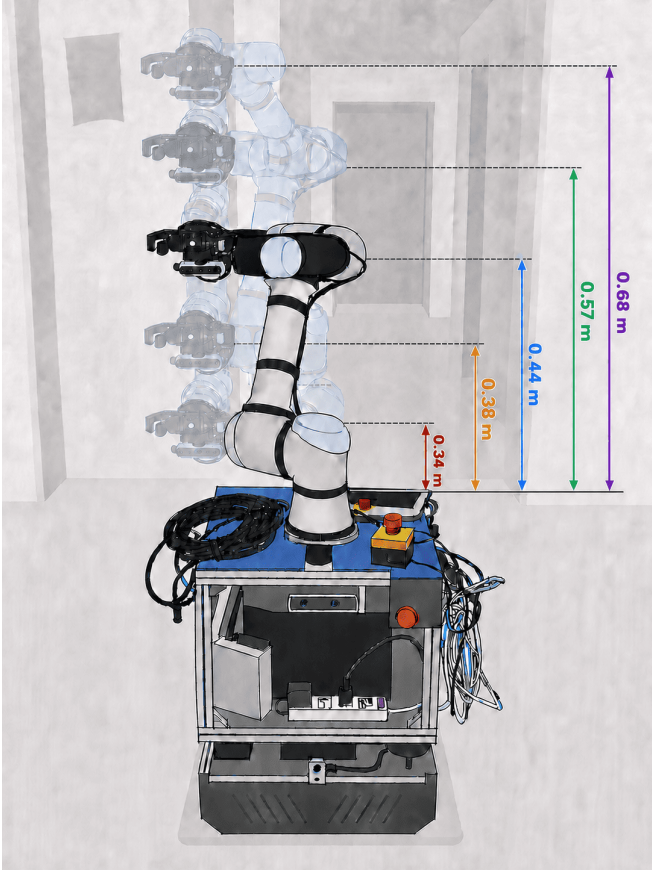


Fig. 5. Initial manipulator configurations used in the end-effector-height variation trials. The gripper center was positioned at 0.34, 0.38, 0.44, 0.57, and 0.68 m relative to the mobile-base reference plane, with 0.44 m representing the nominal configuration.

TABLE III
STAGE-LEVEL PERFORMANCE UNDER END-EFFECTOR HEIGHT VARIATION

Stage	0.34 m	0.38 m	0.44 m	0.57 m	0.68 m
Door search	S	S	S	S	S
Approach	S	S	S	S	S
Handle detection	S	S	S	S	S
3D estimation	S	S	S	S	S
Fine alignment	F	S	S	F	F
Pre-grasp	S	S	S	S	F
Contact / grasp	F	S	S	F	F
End-to-end task	F	F	S	F	F

S = success; F = failure.

quence successfully. The other heights suffered from vertical misalignment, leading to faults in the fine alignment and physical interaction with the handle, which resulted in an overall success rate of 20%.

C. Failure analysis and discussion

The observed failures were grouped into four categories, each highlight a limitation of the approach to the door manipulation pipeline. The first was due to force-limited manipulation, cases where the interaction force was required to exceed the UR7e safety threshold triggering automatic end of the robot's motion. The second issue was the navigation

failure at the largest tested distance of 10.10 m. During this trial, localization drift caused the robot to approach the boundary of the global costmap, preventing from reaching the door and continuing to the subsequent stage. Finally, the third issue was caused by the end-effector height variations leading to misalignment between the gripper and the handle. Despite these failures, the perception pipeline for performing door and handle perception and computing 3D localization were all reliable under the evaluated conditions. The navigation performed successfully to an initial distance of 7.90 m but revealed limitations on recovery on larger approaches. Handle manipulation was the dominant cause of all end-to-end failures, indicating that performing the task with an initial end-effector configuration different to the nominal condition is an area for improvement.

As demonstrated in previous studies on door-opening, robot positioning, compliant interaction, and force regulation during doors opening are critical for success [6]–[9]. From the current characterization of our system, door manipulation is critically dependent on a largely predefined grasp and actuation sequence with no ability to compensate for residual positioning error actively during physical interaction. We believe the addition of a closed-loop pose correction method, alongside an adaptive force control, will be beneficial in reducing geometric and interaction uncertainties while manipulating the door.

It is also worth noting a fourth, more subtle failure mode encountered in the height variation trials, in which the grasp was successful with the end-effector at 0.38 m. Contact-grasp was successfully achieved and the handle was partially actuated; however, the displacement was insufficient to disengage the latch mechanism. At heights of 0.57, 0.68, and 0.34 m, misalignment resulted in grasp failure. This case illustrates the need for closed-loop force or displacement feedback during the actuation phase of the mechanism, rather than feedback based solely on the grasp phase. This distinction highlights the importance of evaluating end to end task completion separately from grasp success.

VI. CONCLUSIONS

In this paper, we presented and evaluated an integrated perception–navigation–manipulation pipeline for semi-autonomous door opening with a mobile manipulator. After manual initialization of the robot pose, the system automatically performs seven stages: door search, approach, handle detection, 3D estimation, fine alignment, pre-grasp, and contact/grasp followed by handle actuation and door opening. Our system was evaluated in 20 trials on the physical robot, under nominal, distance variation, and end-effector height variation conditions. The end-to-end success rate was 80% under nominal conditions, with all stages preceding contact/grasp stage being successful in the 10 trials. The distance variations trials achieved a 80% end-to-end success rate. In contrast, trials under initial end-effector height variation had an end-to-end success rate of only 20% with failures occurring mainly during fine alignment and physical interaction, identifying manipulation as the current

bottleneck under this condition. As future work, we plan to add closed-loop pose correction, adaptive grasp planning, and force-aware manipulation into our system to handle a wider range of door and handle configurations.

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